ECON 300 — APPLIED STATISTICS FOR ECONOMISTS

Fall 2026 · Directed One-on-One Study · Howard University

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Office Hours: By appointment, in person or Zoom
Format: One-on-one directed study

Class Time: Tu/Th 6:15–8:15 AM
First Meeting: Thursday, September 10, 2026
Last Meeting: Thursday, November 19, 2026
Final: 11/19, in class — project presentation

Course Description

This course is a working introduction to applied statistics for economic research. The emphasis is on doing statistics rather than on reciting it: every concept we develop on the whiteboard is implemented in code the same morning, on real data, in a session you could hand to someone else and have them reproduce.

The course is organized around a single arc. We begin with data themselves — what they are, how to describe them, and what a distribution actually tells you. We then build the probability machinery that turns description into inference. From there we spend two weeks on data cleaning, which is where most of the real work of empirical economics happens and which most statistics courses skip. We take a short detour into Bayesian reasoning to make explicit what the frequentist framework leaves implicit. The final month applies all of it to prediction and financial markets, ending in a replication or original project that you present.

Because this is a one-on-one directed study, the pace is yours. If a topic lands quickly we move on; if it does not, we stay. What is not negotiable is the deadline schedule below.

Prerequisites

Intermediate microeconomics and Calculus I. No prior programming experience is assumed, though comfort with a command line will make the first two weeks easier.

Texts

Primary text.

- Mike X Cohen, *Modern Statistics: Intuition, Math, Python, R* (2024). We follow this book closely. It is the only one of the three that gives you the math and the code for both languages on the same page,.

Supplement Text.

- Freedman, Pisani & Purves, *Statistics*, 4th edition. Read this when Cohen's treatment of a foundational idea moves faster than you want. Freedman is the better book on what sampling and probability actually mean; Cohen is the better book on how to compute them.
- Stock & Watson, *Introduction to Econometrics*, Global Edition. This is our bridge from statistics to economics, and the source for the Bayesian block.

Software and Setup

The course is taught in R and Python together, through the Positron IDE. Learning both at once is deliberate: the point is to see that a t-test is a t-test regardless of syntax, and to become the kind of researcher who picks the tool rather than the one their first course happened to teach.

News Subscription — Required Setup

Applied statistics without a sense of what is happening in the world produces technically correct answers to questions nobody asked. Part of this course is training the reflex that turns a news story into a testable empirical question.

You are required to hold an active subscription to *The Wall Street Journal* for the duration of the course. The WSJ student rate is a few dollars a month and is the cheapest of the major business dailies; Howard University Libraries also provides institutional access, which satisfies this requirement at no cost. Set this up before the second session.

Acceptable alternatives, if you prefer one of them:

- Financial Times (student rate)
- Bloomberg (student rate)
- The Economist (student rate)

Deadline. Subscription active and verified by **Tuesday, September 15, 6:15 AM**. Pass/fail.

Structure of a Class Session

Every session is split evenly and the split is strict.

6:15 – 7:15 — Theory. Derivation, intuition, and the assumptions behind the method. Whiteboard and text. You are expected to have done the reading and to interrupt with questions; in a class of one, silence is not an option available to you.

7:15 – 8:15 — Applied code. I write the same method in R and in Python, live, on real data, and you follow along in your own session. This hour is not a lecture — you should be typing. Code from each session is committed to the course repository the same day.

Homework

Problem sets are assigned weekly, drawn from the end-of-chapter exercises in Cohen and their accompanying code notebooks. Each set combines derivation by hand with implementation in both languages. Nine sets are assigned over the term.

Assigned each Thursday after class; due the following **Sunday at 11:59 PM**. Submit as a rendered Quarto document plus the source, committed and pushed to your course repository. A problem set that does not render from a clean clone is not submitted.

News Readings and Discussion

Each week you will pick one news article, a working paper or a data release and we open Tuesday's theory hour by discussing it for ten to fifteen minutes. You must write a one page- memo for this. This should be a bout one paragraph(5-7 sentences) summarizing the text and one paragraph summarizing your thoughts about it such as how you would want to se it in your work your problems with the piece.

Final Project

The course ends with one of two deliverables, your choice, presented in class on November 19.

Option A — Replication. Take a published empirical paper, obtain its data, and reproduce its central result. Then do one thing the authors did not: a robustness check, an alternative specification, an extension to a later period. A replication that fails to reproduce the original is a fine outcome and often a more interesting paper, provided you can show why.

Option B — Original project. Pose an empirical question in prediction or financial markets, assemble the data yourself, and answer it. Scope it small. A narrow question answered cleanly beats a broad one answered vaguely.

Either option is submitted as:

- A paper of roughly 10–15 pages: question, data, method, results, limitations
- A public repository that runs end to end from a clean clone, with a README that a stranger could follow
- A 30-minute in-class presentation and defense of your method choices

The paper and repository are due **Wednesday, November 18, 11:59 PM**.

Course Schedule

The following is the course outline. Topics may shift as we go and I will tell you when they do.

Sess.	Date	Theory hour (6:15–7:15)	Applied code hour (7:15–8:15)	Reading	Hard deadline
ORIENTATION					
1	Thu 9/10	Course orientation; what applied statistics is for	Install Positron; R 4.4 + Python 3.12; renv/uv; Quarto; Git repo init	Cohen Ch. 1; syllabus	Setup verification due Mon 9/14, 11:59 PM; news subscription active by Tue 9/15, 6:15 AM
BLOCK I — DESCRIPTIVE STATISTICS AND DISTRIBUTIONS (9/15 – 9/24)					
2	Tue 9/15	What are data? Types, measurement, accuracy vs. precision; sample vs. population	Positron projects; reading CSV/XLSX in R (readr) and Python (pandas); data dictionaries	Cohen Ch. 2; Freedman Ch. 1–2	—
3	Thu 9/17	Visualizing data: histograms vs. bar plots, box and violin plots, log scaling, color	ggplot2 and matplotlib side by side; one figure, two languages	Cohen Ch. 3	PS 1 due Mon 9/21, 11:59 PM
4	Tue 9/22	Central tendency and dispersion; variance, SD, IQR, heteroscedasticity	Descriptives pipeline; dplyr::summarise vs. pandas .agg	Cohen Ch. 4.1–4.6; Freedman Ch. 3–4	—
5	Thu 9/24	Distributions and the four moments; skew, kurtosis, QQ plots; why simulate	Simulating from named distributions; seeds and reproducibility; QQ plots in both languages	Cohen Ch. 4.7–4.8, Ch. 5	PS 2 due Mon 9/28, 11:59 PM
BLOCK II — PROBABILITY (9/29 – 10/8)					
6	Tue 9/29	What probability is; probability vs. proportion;	Monte Carlo estimation of probabilities; law of large numbers demo	Cohen Ch. 8.1–8.4; Freedman Ch. 13–14	—

Sess.	Date	Theory hour (6:15–7:15)	Applied code hour (7:15–8:15)	Reading	Hard deadline
		analytical vs. empirical probability			
7	Thu 10/1	PMF, PDF, CDF; expected value; the softmax	Plotting densities and CDFs; computing E[X] from simulated and analytic distributions	Cohen Ch. 8.5–8.8	PS 3 due Mon 10/5, 11:59 PM
8	Tue 10/6	Sampling variability, standard error, LLN, and the Central Limit Theorem	Bootstrap sampling distributions; SEM vs. SD in code	Cohen Ch. 9; Freedman Ch. 16–18	—
9	Thu 10/8	Hypotheses, p-values and their misinterpretations; Type I/II error; multiple comparisons	t-tests and confidence intervals in R and Python; reading regression output	Cohen Ch. 10, Ch. 11.1–11.5, Ch. 13	PS 4 due Mon 10/12, 11:59 PM
BLOCK III — DATA CLEANING AND PROCESSING (10/13 – 10/22)					
10	Tue 10/13	Data quality and the cost of getting it wrong; MCAR, MAR, MNAR	Diagnosing every form of missingness, including sentinels (−999, 9999, blank strings)	Cohen Ch. 7.1–7.4; M4 §4.1	Final project topic (1 paragraph) due Mon 10/19, 11:59 PM
11	Thu 10/15	Imputation strategies and when each is right; outliers — Tukey, z-score, iterative z	Imputation menu in pandas/sklearn and R; outlier flagging and trimming	Cohen Ch. 7.5–7.9; M4 §4.2–4.3	PS 5 due Mon 10/19, 11:59 PM
12	Tue 10/20	Transformations: z-score, min-max, log and rank; categorical encoding by cardinality	One-hot vs. target encoding; scaling; pd.cut vs. pd.qcut and their R equivalents	Cohen Ch. 6; M4 §4.4–4.6	—
13	Thu 10/22	Leakage, train/test discipline, and why cleaning belongs in a pipeline	ColumnTransformer (Python) and recipes (R); timed cleaning drill from M4	M4 §4.7 + timed drills	PS 6 due Mon 10/26, 11:59 PM
BLOCK IV — BAYESIAN REASONING (10/27 – 10/29)					
14	Tue 10/27	Conditional probability and Bayes' rule; base rates; priors, likelihood, posterior	Bayes by hand, then by grid approximation in R and Python	Stock & Watson Ch. 2 and Appendix; Cohen Ch. 8.4	—
15	Thu 10/29	Bayesian updating and conjugate priors; Bayesian vs. frequentist interpretation of an interval	Beta-binomial updating; posterior intervals; sensitivity to the prior	Stock & Watson, Bayesian supplement	Project proposal due in class Thu 10/29, 6:15 AM; PS 7 due Mon 11/2, 11:59 PM
BLOCK V — FIELD BLOCK: PREDICTION AND FINANCIAL MARKETS (11/3 – 11/17)					
16	Tue 11/3	Prediction markets as forecasting instruments; prices as probabilities (Election Day)	Pulling live market data; reading an order book	Wolfers & Zitzewitz (2006), "Prediction Markets"	—
17	Thu 11/5	Market microstructure: spreads, depth, liquidity, and where the noise comes from	Polymarket and Kalshi API clients; caching and rate limits; tidying market data	polymarket-arbitrage README; Cohen Ch. 2 review	PS 8 + preliminary results and working repo due Mon 11/9, 11:59 PM
18	Tue 11/10	Correlation and regression applied to market prices; what a coefficient means here	OLS in statsmodels and lm(); residual diagnostics on market data	Cohen Ch. 12, Ch. 15	—

Sess.	Date	Theory hour (6:15–7:15)	Applied code hour (7:15–8:15)	Reading	Hard deadline
19	Thu 11/12	Forecast evaluation: Brier scores, calibration, and the difference between the two	Calibration plots; permutation tests for whether a signal is real	Cohen Ch. 16; Wolfers & Zitzewitz §4	PS 9 due Mon 11/16, 11:59 PM
20	Tue 11/17	Arbitrage detection, power and sample size, and the biases that survive all of it	Backtesting an arbitrage rule; power analysis; writing a reproducible results section	Cohen Ch. 17, Ch. 18, Ch. 19	Final paper + repository due Wed 11/18, 11:59 PM
FINAL					
21	Thu 11/19	FINAL: replication or project presentation, in class	Live walkthrough of code and results; defense of method choices	—	Presentation Thu 11/19, 6:15 AM (hard)

Deadline Summary

Every hard date in the course, in one place.

Deadline	Deliverable	Category
Mon 9/14, 11:59 PM	Software verification: Positron running R and Python, Quarto rendering, Git repo pushed	Pass/fail — gate
Tue 9/15, 6:15 AM	News subscription active (WSJ student rate or approved alternative)	Pass/fail — gate
Mon 9/21, 11:59 PM	Problem Set 1	Homework
Mon 9/28, 11:59 PM	Problem Set 2	Homework
Mon 10/5, 11:59 PM	Problem Set 3	Homework
Mon 10/12, 11:59 PM	Problem Set 4	Homework
Mon 10/19, 11:59 PM	Problem Set 5 — and final project topic, one paragraph	Homework + project
Mon 10/26, 11:59 PM	Problem Set 6	Homework
Thu 10/29, 6:15 AM	Final project proposal: question, data source, replication target, method	Project — hard
Mon 11/2, 11:59 PM	Problem Set 7	Homework
Mon 11/9, 11:59 PM	Problem Set 8 — and preliminary results with working repository	Homework + project
Mon 11/16, 11:59 PM	Problem Set 9	Homework
Wed 11/18, 11:59 PM	Final paper and repository (no extensions)	Project — hard
Thu 11/19, 6:15 AM	Final presentation, in class	Project — hard

Incomplete Policy

Incomplete grades are submitted only where a documented emergency occurs near the end of the term and precludes completion of the final requirements on time. An overwhelming majority of the course requirements must already have been met before the emergency. The written request, with documentation and justification, must be made before final grades are submitted. In this course the student must have:

- Attended the substantial majority of sessions
- Completed at least seven of the nine problem sets
- Submitted an approved final project proposal by the 10/29 deadline

Because the final deliverable is a paper and presentation rather than an exam, an incomplete would apply only where the project could not be finished or presented and the student had otherwise been a full participant. An incomplete is applicable only if completing the project could improve the grade. If granted, the student signs an Incomplete Contract specifying the submission date.

Americans with Disabilities Statement

Howard University is committed to creating an accessible, inclusive, and safe learning environment for all students and providing equal access to students with documented disabilities. Students seeking reasonable accommodation must first register with the Office of Student Services (OSS). There you can engage in a confidential conversation about the process for requesting reasonable accommodations in the classroom and clinical settings, which the Office of Student Services (OSS) determines. Accommodations must be requested each semester. Accommodations are not provided retroactively. If you want to request accommodations, please contact OSS via email at oss.disabilityservices@howard.edu or visit https://howard.edu/disability-services.

Statement on Sex and Gender-Based Discrimination, Harassment, and Violence

Howard University's Policy Prohibiting Sex and Gender-Based Discrimination, Sexual Misconduct and Retaliation (the Title IX Policy) prohibits discrimination, harassment, and violence based on sex, gender, gender expression, gender identity, sexual orientation, pregnancy, or marital status. With the exception of certain employees designated as confidential, all Howard University employees — including all faculty members — are required to report any information they receive regarding known or suspected prohibited conduct under the Title IX Policy to the Title IX Office (TitleIX@howard.edu or 202-806-2550), regardless of how they learn of it. For confidential support and assistance, you may contact the Interpersonal Violence Prevention Program (202-836-1401) or the University Counseling Service (202-806-7540). To learn more about your rights, resources, and options for reporting and/or seeking confidential support services, including additional confidential resources both on and off campus, visit titleix.howard.edu.

Academic Integrity and the Use of AI

You may use AI assistants for this course, and you should — they are part of how empirical work is now done, and Cohen's own text includes a section on it. Two rules. First, you must be able to explain every line of code you submit; if you cannot defend it in the applied hour, it does not count as yours. Second, disclose it: a short note at the end of each problem set saying what you used and for what. Undisclosed use, or code you cannot explain, is an integrity violation under Howard's Student Code of Conduct.